

CSE323 — Software Engineering

Term Project: The Full Engineering Cycle

Evaluation Criteria & Grading Rubric - Spring 2026

01 GROUP FORMATION & PROJECT SCOPE

Rule	Details
Group Size	Minimum 3 members — Maximum 5 members
Sub-system Choice	Each group selects one sub-system for example: • Customer Ordering System • Kitchen Display System • Management Analytics System you can choose your own sub system those are only examples
AI Disclosure	All AI-assisted code must be disclosed; prompts used must appear in the appendix

02 DELIVERABLES

Groups must submit **ALL** of the following deliverables by **18 / 05 / 2026**.

#	Deliverable	Format
D1	Project Registration Form Group members, roles, and chosen sub-system	Google Form
D2	Requirements Report Actor classification, traceability heatmap, persona discovery (≥ 5 edge cases)	PDF (max 6 points)
D3	Design Specification Gherkin scripts, UML diagrams (SSD + Activity), API contracts, QA audit log	PDF (max 10 points)
D4	Validation Report Test pyramid results, Playwright scripts, Verification vs Validation statement	PDF (max 8 points) Note that you will deliver all parts (D2,D3,D4) in one pdf file
D5	Implementation Package Source code + TDP evidence (failing → passing tests), vertical slice demo (screen recording ≤ 5 min)	All project files+ GitHub link + video link
D6	Final Presentation Slide Deck Submitted before the demo session	PPTX or PDF

03 GRADING RUBRIC (Total: 30 Marks)

Each phase is assessed holistically across three performance levels: **Excellent** (full marks), **Satisfactory** (partial marks), and **Insufficient** (zero). Partial marks within each band are awarded at the marker's discretion.

Phase 1 — Requirement Discovery & Traceability

Criterion	Excellent (Full Marks)	Satisfactory (Partial)	Insufficient (Zero)
Actor Classification	All three actor types (Primary, Supporting, Offstage) correctly identified with clear rationale	Two actor types defined; minor omissions or unclear rationale	One or no actor types addressed
Traceability Heatmap	Complete matrix with zero orphaned requirements every feature are implemented and justified	Matrix present but 1–2 orphaned requirements or missing justifications	No matrix, or matrix is largely incomplete
Persona Discovery	At least 5 hidden requirements or edge cases uncovered via AI user avatar with clear reasoning	3–4 edge cases uncovered; some lack depth or justification	Fewer than 3 edge cases or persona exercise not conducted

Phase 2 — Design & Specification

Criterion	Excellent (Full Marks)	Satisfactory (Partial)	Insufficient (Zero)
Gherkin Scripts	All core user stories translated to well-formed Given/When/Then with scenario outlines where appropriate	Most stories covered; minor syntax issues or missing edge scenarios	Fewer than half of stories covered, or Gherkin syntax incorrect throughout
QA Refinement Loop	All vague adjectives replaced with measurable metrics; senior QA audit log provided to eliminate unquantifiable adjectives like "fast" or "secure" and replace them with measurable technical metrics.	Most adjectives quantified; 1–2 remain vague	Vague terms remain unquantified; no audit log
UML Diagrams	Correct SSD (System Sequence Diagrams) (happy + failure paths) and Activity diagram with code decision points; notation accurate	Diagrams present but missing failure path or decision points; minor notation errors	Diagrams missing or fundamentally incorrect
API Contracts & Information Hiding	Clear interface definitions; internal stack fully hidden; contracts enable independent development	Interfaces defined but some internal details leak or contracts are incomplete	No API contracts or no separation of interface from implementation

Phase 3 — Test-Driven Implementation

Criterion	Excellent (Full Marks)	Satisfactory (Partial)	Insufficient (Zero)
Failing Test First	Failing unit tests written before implementation; tests establish a clear mathematical boundary	Tests written but some produced after code, or boundaries poorly defined	No failing-first evidence, or tests absent
Edge Case Cage (Padlocks)	Boundary, threshold, and extreme constraints all addressed; AI hallucination blocked for all cases	At least two padlock types applied; some edge cases unaddressed	Padlocks absent or only one constraint type applied
TDP Iteration	Iterative prompting documented; final code demonstrably fits the test boundary; prompt appendix complete	Some iteration shown; minor misalignment between test and final logic	No iteration documented; code does not match test boundary
Vertical Slicing	UI / Logic / DB delivered as one working vertical slice; failure resilience demonstrated	Vertical slice partially working; one layer missing or not integrated	No vertical slice; monolithic or incomplete implementation

Phase 4 — Validation & Pipeline Engineering

Criterion	Excellent (Full Marks)	Satisfactory (Partial)	Insufficient (Zero)
Testing Pyramid	70% Unit / 20% Integration / 10% E2E ratio achieved and evidenced with a test report	Ratio attempted but deviates by more than 15% in any layer	Testing pyramid not followed or no test report provided
Playwright Automation	Gherkin scenarios fully converted to executable Playwright scripts using the Page Object Model	Scripts present but POM not applied, or not all scenarios covered	No Playwright scripts, or scripts not linked to Gherkin
Verification vs Validation Statement	Clear written statement distinguishing what was verified (works) vs validated (right problem), with evidence	Statement present but conflates the two concepts or lacks evidence	No distinction made, or statement absent